Trace ($\mathrm{tr}()$)

- The trace of a scalar is the scalar itself: $a = \mathrm{tr}(a)$
- Let ABC be a matrix moltiplication that results in a scalar, then $ABC = \mathrm{tr}(ABC)$
- **Cyclic property:** $\mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$
- **Linearity:** $\mathrm{tr}(A+B) = \mathrm{tr}(A) + \mathrm{tr}(B)$
- Trace and expected value are interchangeable $\mathbb{E}[ABC] = \mathbb{E}[\mathrm{tr}(ABC)] = \mathrm{tr}(\mathbb{E}[ABC])$

Complex Gaussian Distribution ($\mathcal{N}_{\mathbb{C}}$)

- $x \sim \mathcal{N}_{\mathbb{C}}(\mu, \sigma^2) \longrightarrow f_x(x) = \dfrac{1}{\pi\sigma^2} \exp\left\{\dfrac{-|x-\mu|^2}{\sigma^2}\right\}$ for scalars $h(x) = \log_2(\pi e \sigma^2)$
- $x \sim \mathcal{N}_{\mathbb{C}}(\mu, R) \longrightarrow \dfrac{1}{\det(\pi R)} \exp\left\{-(x-\mu)^* R^{-1}(x-\mu)\right\}$ for vectors $h(x) = \log_2(\pi e R)$

1) A primer on information theory and MMSE estimation

1.1) Information Content

Information is defined as

$$i(A) = \log_2 \left(\frac{1}{P[A]}\right) [bit]$$

Given a discrete r.v. with alphabet $\mathcal{A}_x = \{a_1, \dots, a_M\}$ it is possible to describe a **information function**

$$i_x : \mathcal{A}_x \to \mathbb{R},\ i_x(a) = i(x = a) = \log_2 \frac{1}{p_x(a)}$$

Entropy

From here we can first define the entropy:

Definition 1 (Entropy)**.**

Let x be a discrete RV with PMF $p_x(\cdot)$. It's entropy is defined as

$$H(x) = -\sum_x p_x(x) \log_2 p_x(x) = -\mathbb{E}[\log_2 p_x(x)]$$

Since we will be dealing with MIMO we can represent a vector of PMF as $\mathbf{x}$ and the **joint entropy** is

$$H(\mathbf{x}) = -\mathbb{E}[\log_2 p_{\mathbf{x}}(\mathbf{x})]$$

where $p_{\mathbf{x}}$ is the joint PMF of x_i RVs.

This value quantifies the uncertainty associated with x, that is, the larger the entropy, the more unpredictable.

Theorem 2 (Entropy Properties)**.**

- *$H \geq 0$ is non negative. It is zero when x is deterministic*
- *The PMF that maximizes H is the uniform distribution*
- *A shifted rv has the same entropy: $H(x + c) = H(x)$ with $c \in \mathbb{C}$*
- *$H(x|y) \leq H(x)$*
- *Let x have cardinality M, then $H(x) \leq \log_2 M$*
- *Let x_i be independent, then $H(\mathbf{x}) = \sum_i H(x_i)$*

Moreover we can also define the **conditional entropy**

$$H(x|y) = -\sum_x \sum_y p_{xy}(x, y) \log_2 p_{x|y}(x|y)$$

if x is deterministic of y this yields 0.
This is related to the joint entropy as

$$H(x, y) = H(x) + H(y|x)$$

and in the general sense (**chain rule**)

$$H(x_0,\dots,x_{N-1}) = H(\mathbf{x}) = \sum_{n=0}^{N-1} H(x_n|x_0,..,x_{n-1}) = \sum_{n=0}^{N-1} H(x_n|\mathbf{x})$$

Another quantity can be defined. This is **devoid of operational meaning**.

Definition 3 (Differential Entropy)**.**

Let $f_x(\cdot)$ be the PDF of x, its differential entropy is

$$h(x) = -\int_{\text{support of x}} f_x(x)\log_2 f(x)dx = -\mathbb{E}[\log_2 f_x(x)]$$

Remark (Differential Entropy Can Be Negative)**.**

The differential entropy can be negative.
Proof (example 1.4):
Let $x \sim U[0,b]$, then

$$h(x) = -\int_0^b \frac{1}{b}\log_2\frac{1}{b}dx = (\frac{1}{b}\log_2\frac{1}{b})|_0^b = \log_2 b$$

which is clearly negative if $b < 1$

TODO Entropy Rate

Definition 4 (Relative Entropy)**.**

Consider two PMFs $p(\cdot)$. $q(\cdot)$.If $q(\cdot) \neq 0 \in$ support of $p(\cdot)$, the their relative entropy is

$$D(p||q) = \sum_x p(x)\log_2\frac{p(x)}{q(x)} = \mathbb{E}\left[\log_2\frac{p(x)}{q(x)}\right]$$

Note that his is not symmetric $D(q||p) \neq D(p||q)$. This is interpreted as a similarity measure between q and p, and it is 0 iff $p = q \; \forall x$.
Moreover for PDFs:

$$D(f||g) = \int f(x)\log_2\frac{f(x)}{g(x)}dx$$

Examples

First an example on entropy:

Example 1.2/1.3 (Entropy of bernoulli and M equiprobable points)**.**

Here are some examples:

1. *Let x be bernoulli, that is:*

$$x = \begin{cases} 0 & \text{with probability } p \\ 1 & \text{with probability } 1-p \end{cases}$$

The entropy is therefore

$$H(x) = -p\log_2 p - (1-p)\log_2(1-p)$$

And clearly it this is maximized when $p = 1/2$.

2. *M equiprobable points, that is $x \sim U[1/M]$*
   The entropy is

$$H(x) = -\sum_{m=0}^{M-1} \frac{1}{M}\log_2\frac{1}{M} = -\log_2\frac{1}{M} = \log_2 M$$

Now a fundamental result for differential entropy:

Example 1.5/1.6 (Differential Entropy of complex Gaussian vector/scalar)**.**

Let $x \sim \mathcal{N}_{\mathbb{C}}(\mu, \mathbf{R})$, then

$$\begin{aligned}
h(x) &= -\mathbb{E}[\log_2 f_x(x)] = -\mathbb{E}\left[\log_2 \frac{1}{\det(\pi R)} \exp\left\{-(x-\mu)^* R^{-1}(x-\mu)\right\}\right] \\
&= \underbrace{\log_2 \det(\pi R)}_{a} + \mathbb{E}[(x-\mu)^* R^{-1}(x-\mu)]\log_2 e \\
&= a + \mathrm{tr}(\mathbb{E}[R^{-1}(x-\mu)(x-\mu)^*])\log_2 e \\
&= a + \mathrm{tr}(R^{-1}\mathbb{E}[(x-\mu)(x-\mu)^*])\log_2 e \\
&= a + \mathrm{tr}(I)\log_2 e \\
&\boxed{h(x) = \log_2 \det(\pi e R)}
\end{aligned}$$

This is a bit cumbersome, so let's go step by step:

- *The first two lines just state the problem, and use known identities and properties*
- *The third line uses three tricks: the trace of a scalar ($\mathbb{E}$) is equal to it's trace. I recalled that $\log_2 e^x = x\log_2 e$. Moreover notice that $(x-\mu)^* R^{-1}(x-\mu)$ is a scalar, so by using the properties of trace (cyclic and $E[\mathrm{tr}(A)] = \mathrm{tr}(E[A])$)*
- *In the fourth line I just moved R^{-1} outside of $\mathbb{E}$*
- *In the fifth lune I have that $\mathbb{E}[\cdot] = R$ and thus $R^{-1}R = I$*
- *Finally recall that* $\exp\{\mathrm{tr}(A)\} = \det(\exp\{A\})$ *and that* $\mathrm{tr}(I) = n$*, then:* $\mathrm{tr}(I)\log_2 e = n\log_2 e = log_2 e^n = \log_2 e^{tr(I)} = \log_2 \det(\mathrm{tr}(e^I))$ *and then recall $e^I = eI$ so finally* $a + \log_2 \det(eI) = \log_2 \det(\pi R e I) = \log_2 \det(\pi e R)$

Now for the gaussian scalar:
Let $x \sim \mathcal{N}_{\mathbb{C}}(\mu, \sigma^2)$ we have

$$h(x) = \mathbb{E}\left[\frac{|x-\mu|^2}{\sigma^2}\log_2 e + \log_2(\pi\sigma^2)\right] = \log_2 \pi e \sigma^2$$

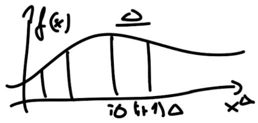
This is more straight-forward as:
$\mathbb{E}\left[|x-\mu|^2\right] = \sigma^2$ *and thus the first term equals to* $\log_2 e$ *and the second just* $\log_2 \pi\sigma^2$ *and thus it is easily proven.*

Finally, a more mathematical concept that relates (or actually shows no relation) between entropy and discrete entropy:

Example ($H(x)$ cannot be approached by discretizing f_x)**.**

The entropy of a b-bit uantization is approximately $h(x) + b$, however this diverges as $b \to \infty$.
From the mean value thm we have that

$$\exists x_i \in [i\Delta, (i+1)\Delta] : f(x_i) \cdot \Delta = \int_{i\Delta}^{(i+1)\Delta} f(x)dx$$



Bad Graphical representation

Now consider $x^\Delta = x_i \in [i\Delta, (i+1)\Delta)$ by the mean value thm we have the discretized PMF $P[x_i] = \int f(x)dx = f(x_i)\Delta = p_i$. What happened is that by the mean val thm and by a small Δ we approximated $f \approx p$. This is exactly how integrals are computed, so this should work, right?

The entropy of this quantized rv is found by

$$\begin{aligned} H(x^\Delta) &= -\sum_i p_i \log_2 p_i \\ &= -\sum \Delta f(x_i) \log_2(\Delta f(x_i)) \\ &= \sum \Delta f(x_i) \left(\log_2 \Delta + \log_2 f(x_i)\right) \\ &\text{notice how } \sum_i \Delta f(x_i) \log_2 \Delta = \log_2 \Delta \\ &= \underbrace{-\sum \Delta f(x_i) \log_2 f(x_i)}_{h(x)} - \log_2 \Delta \\ H(x^\Delta) &= h(x) - \log_2 \Delta \to h(x) = H(x^\Delta) + \log_2 \Delta \end{aligned}$$

Where $\sum \Delta f(x_i) = 1$ by definiton of CDF.
This holds for a uniform rv $x \sim U(0,1)$ as $\Delta = 2^{-n}$ and thus $H(x^\Delta) = n$ but this doesn't hold for a gaussian rv as it needs more than n bits: $x \sim N(0, \sigma^2)$, $\Delta = 2^{-n}$ $H(x^\Delta) = \log(\pi e \sigma^2) + n$, so information is lost and the approximation is not valid.

□

Mutual Information

Definition 5 (Mutual Information)**.**

The mutual information between two RVs is the reduction in uncertainty about s once y is known and vice versa.

$$\begin{aligned} I(s; y) &= H(s) - H(s|y) \\ &= H(y) - H(y|x) \end{aligned}$$

Or also the relative entropy of joint PMF and the marginals

$$
\begin{aligned}
I(s;y) &= D(p_{sy}||p_s p_y) \\
&= \sum_x \sum_y p_{sy}(s,y) \log_2 \frac{p_{sy}(s,y)}{p_s(s)p_y(y)} \\
&= \sum_x \sum_y p_{sy}(s,y) \log_2 \frac{p_{y|s}(y|s)}{p_y(y)}
\end{aligned}
$$

In the continuous case the same holds

$$
\begin{aligned}
I(s;y) &= h(s) - h(s|y) \\
&= h(y) - h(y|x) \\
&= D(f_{sy}||f_s f_y)
\end{aligned}
$$

And it also expands to vectors (just replace s, y with $\mathbf{s}, \mathbf{y}$)

In contrast with differential entropies, $I(s,y)$ can be obtained by the limit of discretized version of s, y, albeit the differential entropies diverge, their difference is still well behaved.

Moreover the mutual information satisfies the **chain rule:**

$$
I(x_0, \ldots, x_{N-1}; y) = \sum_{n=0}^{N-1} I(x_n; y|x_0, \ldots, x_{N-1})
$$

Examples

From the definition, we can show all equivalencies

Exercise (Some equivalencies proved)**.**

Show that $H(s) - H(s|y) = H(y) - H(y|x)$.
Recall that $H(s|y) = H(s,y) - H(y)$ *and* $H(s,y) = H(y,s)$. *Rewrite*

$$
H(s) - H(s|y) = H(s) + H(y) - H(s,y) = H(y) - H(y|s)
$$

□

Now show $H(s) - H(s|y) = D(p_{sy}||p_s p_y)$
Recall that $H(s|y) = -\sum_s \sum_y p(s,y) \log_2 \frac{p(s,y)}{p(s)p(y)}$ *by definition and that*
Recall that the sum of the joint PMF over one is the marginal of the other: $\sum_y p(s,y) = p(s)$ *and we can rewrite*

$$
H(s) = -\sum_s p(s) \log_2 p(s) = -\sum_s \left(\sum_y p(s,y)\right) \log_2 (p(s) = -\sum_s \sum_y p(s,y) \log_2 p(s)
$$

Now rewrite

$$
H(s) - H(s|y) = \sum_s \sum_y p(s,y) \left[\log_2 p(s|y) - \log_2 p(s)\right] = -\sum_{s,y} p(s,y) \log_2 \frac{p(s,y)}{p(s)p(y)}
$$

Where in the last step we noticed $p(s|y) = p(s,y)/p(y)$

□

Now, let's continue with the discretization example:

Exercise (Discretization of X and Y can lend to mutual information calculation)**.**

By recalling the result of [^98221d](#^98221d) *we can also find* $H(X^\Delta|Y^\Delta)$.
In fact, by applying the mean val thm in 2D we have that

$$P_{x,y} = \iint f_{xy}(x,y)\,dxdy = \Delta_x\Delta_y f_{x,y} = p_{x,y}$$

And thus

$$p_{x|y} = \frac{p_{x,y}}{p_y} = \frac{\Delta_x\Delta_y f_{x,y}}{\Delta_y f_y} = \Delta_x \frac{f_{x,y}}{f_y} = \Delta_x f_{x|y}$$

And as before

$$\begin{aligned} H(X^\Delta|Y^\Delta) &= -\sum_x\sum_y p(x,y)\mathrm{log}_2 p(x|y) \\ &= -\sum_x\sum_y \Delta_x\Delta_y f(x,y)\,(\mathrm{log}_2\Delta_x - \mathrm{log}_2 f(x|y)) \\ &= -\mathrm{log}_2\Delta_x + \sum_x\sum_y \Delta_x\Delta_y(x,y)\mathrm{log}_2 f(x|y) \\ &= h(x|y) - \mathrm{log}_2\Delta_x \end{aligned}$$

Then just compare:

$$I(x;y) = H(X^\Delta) - H(X^\Delta|Y^\Delta) = h(x) - \mathrm{log}_2\Delta_x - (h(x|y) - \mathrm{log}_2\Delta_x) = h(x) - h(x|y)$$

□

Finally, another key result is the gaussian mutual information

Example 1.7 (Gaussian Mutual Information for a Complex Scalar)**.**

Let $y = \sqrt{\rho}s + z$ *with* $s, z \sim \mathcal{N}_\mathbb{C}(0,1)$ *independent. Notice* $y \sim \mathcal{N}_\mathbb{C}(0, 1+\rho)$ *and* $y|s \sim \mathcal{N}_\mathbb{C}(\sqrt{\rho}s, 1)$ *we have*

$$\begin{aligned} I(\rho) &= I(s; \sqrt{\rho}s + z) = h(\sqrt{\rho}s + z) - h(\sqrt{\rho}s + z|s) \\ &= \log_2(\pi e(1+\rho)) - h(z) = \log_2(\pi e(1+\rho)) - \log_2(\pi e) \\ &= \log_2(1+\rho) \end{aligned}$$

Why can we say this about y *and* $y|s$

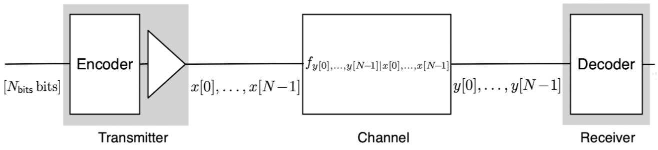
- *for* y *it is a sum of two independent gaussians so* $y \sim \mathcal{N}_\mathbb{C}(0+0, (\sqrt{\rho})^2 + 1)$
- *for* $y|s$*, since* s *is known, it is just a shift of* $\sqrt{\rho}s$
  This can be read as: If z *is the noise, then* ρ *is the SNR and* $I(\rho)$ *is the mutual information of an AWGN signal*

Example 1.13 (Gaussian Mutual Information for a Complex Vector)**.**

TODO

1.2) Reliable Communication

Let's start by explaining the communication link:



Basic Communication Link

- The encoder parses N_b bits into messages. These bits are IID and bernoulli.
- Each message is a collection of N_b bits and therefore there exist 2^{N_b} messages.
- Each message is mapped into N unit power complex symbols $s[0], \ldots, s[N-1]$
- Each symbol is then amplified $x[0], \ldots, x[N-1]$
- The channel randomly modifies the signal. The **channel is known** if the conditional probability $f_{y|x}$ ($f_{y[0],\ldots y[N-1]|x[0,\ldots,x[N-1]]}$) is known. Since the power amplification is known, this directly allows to find $f_{y|s}$.
- The decoder maps the observed values to the correct (ideally) codeword

Remark.

Notice how $N_b \neq N$. N_b is the number of bits to transmit, while N the number of codewords (symbols).

Although we are in discrete time, and signals are continuous in time, this approximation is valid due to shannon's sampling theorem. Moreover the bits to transmit are IID and therefore already underwent necessary redundency reductions.

For notation we introduce bold overlined symbols to denote **time-domain sequences**

$$\overline{\mathbf{s}} = \begin{bmatrix} s[0] \\ \ldots \\ s[N-1] \end{bmatrix}$$

(For obsidian notes I introduce the command \tds{})

$$\overline{\textbf{input}}$$

Therefore, with N_b IID bits, 2^{N_b}equiprobable codewords, the mutual information between input and output is:

$$I(\overline{\mathbf{s}}; \overline{\mathbf{y}}) = \mathbb{E}\left[\log_2 \frac{f_{\overline{\mathbf{y}}|\overline{\mathbf{s}}}(\overline{\mathbf{y}}|\overline{\mathbf{s}})}{f_{\overline{\mathbf{y}}}(\overline{\mathbf{y}})}\right]$$

$$\stackrel{\text{bayes}}{=} \mathbf{E}\left[\log_2 \frac{f_{\overline{\mathbf{y}}|\overline{\mathbf{s}}}(\overline{\mathbf{y}}|\overline{\mathbf{s}})}{\frac{1}{2^{N_b}} \sum_{m=0}^{2^{N_b}-1} f_{\overline{\mathbf{y}}|\overline{\mathbf{s}}}(\overline{\mathbf{y}}|\overline{s}_m)}\right]$$

Definition 6 (Memoryless)**.**

We call a channel memoryless, that is, if the output of symbol n depends only of the input at symbol n, if the channel law factors as

$$f_{\overline{\mathbf{y}}|\overline{\mathbf{s}}}(\cdot) = \prod_{n=0}^{N-1} I(s[n]; y[n])$$

Then, it is possible to have statistically independent codewords and entries and the mutual information becomes

$$I(\overline{\mathbf{s}};\overline{\mathbf{y}}) = \sum_{n=0}^{N-1} I(s[n];y[n])$$

with

$$I(s[n];y[n]) = \mathbb{E}\left[\log_2 \frac{f_{y[n]|s[n]}(y[n]|s[n])}{f_{y[n]}(y[n])}\right]$$

If we add also stationary (time-invariant), then we can drop index n and write

$$I(s;y) = \mathbb{E}\left[\log_2 \frac{f_{y|s}(y|s)}{f_y(y)}\right] = \mathbb{E}\left[\log_2 \frac{f_{y|s}(y|s)}{\sum_{m=0}^{M-1} f_{y|s}(y|s_m)p_m}\right]$$

Where it applies to an M-point constellation.

Example 1.16 (Memoryless channel with AWGN)**.**

We can write AWGN as

$$y[n] = \sqrt{\rho}s[n] + z[n]$$

With $z[n_i]$ IID gaussian $z \sim \mathcal{N}_{\mathbb{C}}(0,1)$.
By conditioning on $y|s$ we can see $\sqrt{\rho}s[n]$ as a constant and thus $y|s \sim \mathcal{N}_{\mathbb{C}}\sqrt{\rho}s, 1$
By recalling that for a complex gaussian the PDF is

$$f_z(z) = \frac{1}{\pi\sigma^2} e^{-|z-\mu|^2/\sigma^2}$$

Then, the channel law is:

$$f_{y|s}(y|s) = \frac{1}{\pi} e^{-|y-\sqrt{\rho}s|^2}$$

1.3) Capacity

Capacity is the **maximum rate** at which information can be transmitted over a channel with **arbitrarily small error probability** if there exists an adeguate code, that is $p_e \to 0 \iff N \to \infty$. If this limit is exceeded, the error probability rises rapidly.

Therefore, since infinitely long signals aren't physically realizable we have that

$$\frac{R}{B} \leq C$$

where R/B is the spectral efficiency $\left(\frac{bit/s}{Hz}\right)$.

Definition 7 (Capacity)**.**

Shannon studied **stationary, ergodic and memoryless** channels, that is:

$$C_{\text{stationary, ergodic, memoryless}} = \max I(s;y)$$

By relaxing the memoryless condition

$$C_{\text{ergodic, stationary}} = \max \lim_{N\to\infty} \frac{1}{N} I(\bar{\mathbf{s}};\bar{\mathbf{y}})$$

This if the channel is information stable. And the symbols $s[n]$ are zero mean.

In an arbitrary channel we have the following probability to make an error:

$$p_e = \sum_{m=0}^{2^{N-b}-1} P[\hat{w} \neq m | w = m] P[w = m] \overset{\text{equiprobable}}{=} \frac{1}{2^{N_b}} \sum_{m=0}^{2^{N-b}-1} P[\hat{w} \neq m | w = m]$$

Definition 8 (Information Stable and Information Density)**.**

The information density is the quantity whose expected value is the mutual information

$$i(\bar{\mathbf{s}};\bar{\mathbf{y}}) = \log_2 \frac{f_{\bar{\mathbf{s}},\bar{\mathbf{y}}}(\bar{\mathbf{s}},\bar{\mathbf{y}})}{f_{\bar{\mathbf{s}}}(\bar{\mathbf{s}}) f_{\bar{\mathbf{y}}}(\bar{\mathbf{y}})} \to I(\bar{\mathbf{s}};\bar{\mathbf{y}}) = \mathbb{E}[i(\bar{\mathbf{s}};\bar{\mathbf{y}})]$$

The channel is **information stable** if

$$\begin{aligned} \lim_{N\to\infty} \frac{1}{N} i(\bar{\mathbf{s}};\bar{\mathbf{y}}) &= \lim_{N\to\infty} \frac{1}{N} \mathbb{E}[i(\bar{\mathbf{s}};\bar{\mathbf{y}})] \\ &= \lim_{N\to\infty} \frac{1}{N} I(\bar{\mathbf{s}};\bar{\mathbf{y}}) \end{aligned}$$

This also shows that the information density doesn't deviate asymptotically from the mutual information.

1.4) Coding and Decoding

TODO

1.5) MMSE Estimation

Estimation refers to the study on how and with what accuracy one can infer the value sent by the one obtained. We always referr to the AWGN model:

$$y = \sqrt{\rho} s + z$$

We have the following information available at the receiver side:

- Obtained y
- Posteriori probability $f_{s|y}(\cdot)$
- Likelihood function $f_{y|s}(\cdot)$
- Prior probability (sometimes) $f_s(\cdot)$
- Fidelity Criterion

The criterion of this chapter is the Minimum Mean Square Error (MMSE), which is the solution that minimizes $\mathbb{E}[|s - \hat{s}(y)|^2]$.

Definition 9 (MMSE Estimator)**.**

The MMSE estimator is

$$\hat{s}(y) = \mathbb{E}[s|y]$$

So the **coditional mean estimator** is

$$MMSE(\rho) = \mathbb{E}\left[(s - |\mathbb{E}[s|y]|)^2\right]$$

The MMSE is unbiased (over all randomness) since $\mathbb{E}[\hat{s}(y)] = \mathbb{E}[\mathbb{E}[s|y]] = \mathbb{E}[s]$. A quick proof is provided

$$\mathbb{E}[\mathbb{E}[s|y]] = \int \left(\int s \cdot p(s|y)ds\right)p(y)dy = \int s\left(\int p(s|y)p(y)dy\right)ds = \int s \cdot p(s)ds = \mathbb{E}[s]$$

But it is biased (for individual realizations $|s$) as it might be that $E[\hat{s}(\sqrt{\rho}s + z)|s] \neq s$.

The MMSE can also be generalized to vectors, in fact:

Definition 10 (MMSE Estimator for Vectors)**.**

Let

$$\mathbf{y} = \sqrt{\rho}\mathbf{A}\mathbf{s} + \mathbf{z}$$

Where A is fixed, s and z are independent and $z \sim \mathcal{N}_{\mathbb{C}}(0, \mathbf{R})$, then **the conditional mean estimator is**

$$\hat{\mathbf{s}}(\mathbf{y}) = \mathbb{E}[\mathbf{s}|\mathbf{y}]$$

and the MMSE is

$$\mathbf{E} = \mathbb{E}\left[(s - \hat{s}(y))(s - \hat{s}(y))^*\right]$$

which is the covariance matrix and the j-th diagonal gives the error associated with estimator j-th entry of s.

Examples

The following two examples will show the bias of MMSE in individual realizations

Example 1.27 (MMSE of Complex Gaussian Scalar)**.**

Let $y = \sqrt{\rho}s + z$ and $s \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, then by definition:

$$f_s(s) = \frac{1}{\pi}e^{-|s|^2}$$

Moreover recall that $y|s \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}s, 1)$ and thus

$$f_{y|s}(y|s) = \frac{1}{\pi}e^{-|y-\sqrt{\rho}s|^2}$$

Now find the MMSE estimator $\hat{s}$:

$$\begin{aligned}\hat{s}(y) = \mathbb{E}[s|y] = \int s f_{s|y}(s|y)ds = \int \frac{s f_{y|s}(y|s)f_s(s)}{f_y(y)}ds = \frac{\int s f_{y|s}(y|s)f_s(s)ds}{f_y(y)} \\ = \frac{\int s f_{y|s}(y|s)f_s(s)ds}{\int f_{y|s}(s)f_s(s)ds} = \boxed{\frac{\int s e^{-|y-\sqrt{\rho}s|^2}f_s(s)ds}{\int e^{-|y-\sqrt{\rho}s|^2}f_s(s)ds}} \\ = \frac{\int s e^{-|y-\sqrt{\rho}s|^2}e^{-|s|^2}ds}{\int e^{-|y-\sqrt{\rho}s|^2}e^{-|s|^2}ds}\end{aligned}$$

This is a bit cumbersome, let's see:

- *In the first line we applied the definition, then Bayes, and then we recognized that* $f_y(y)$ *does not depend on* s.
- *The second line used the marginal definition for* $f_y(y)$ *and then used the know equation fo* $f_{y|s}(y|s)$
- *The third was obtained by using* $f_s(s)$ *of this specific problem.*

Now look at the exponent (ignore -1 multiplier):

$$|y-\sqrt{\rho}s|^2-|s|^2=|y|^2+(1-\rho)|s|^2-\sqrt{\rho}(y^*s+ys^*)$$

This can be rewritten as

$$\frac{|y|^2}{1+\rho}+\left|\sqrt{1+\rho}s-\sqrt{\frac{\rho}{1+\rho}}y\right|^2$$

Since the first term ($\exp\left\{|y|^2/1+\rho\right\}$) *is not dependent on s it can be moved out of both integrals and simplified, moreover the square can collect* $\sqrt{1+\rho}$ *and become*

$$(1+\rho)\left|s-\frac{\sqrt{\rho}}{1+\rho}y\right|^2=\left|s-\frac{\sqrt{\rho}}{1+\rho}y\right|^2\frac{1}{\frac{1}{1+\rho}}$$

Now multiply both sides by $1/\pi(1/(1+\rho))$ *and therefore the exponents become*

$$\frac{1}{\pi\left(\frac{1}{1+\rho}\right)}e^{\frac{-|s-\frac{\sqrt{\rho}}{1+\rho}y|^2}{1/1+\rho}}$$

which is the PDF of $s|y\sim\mathcal{N}_{\mathbb{C}}(\frac{\sqrt{\rho}}{1+\rho}y,\frac{1}{1+\rho})$.
Therefore

$$\frac{\int se^{-|y-\sqrt{\rho}s|^2}e^{-|s|^2}ds}{\int e^{-|y-\sqrt{\rho}s|^2}e^{-|s|^2}ds}=\frac{\int sf_{s|y}(s|y)ds}{\int f_{s|y}(s|y)ds}$$

The denominator is clearly 1 while the numerator is the expected value of $s|y$ *and thus we have*

$$\hat{s}(y)=\frac{\sqrt{\rho}}{1+\rho}y$$

Now

$$MMSE(\rho)=\mathbb{E}\left[\left|s-\frac{\sqrt{\rho}}{1+\rho}y\right|^2\right]$$

Compute the square

$$|s|^2+\frac{\rho}{(1+\rho)^2}|y|^2-(\frac{\sqrt{\rho}}{1+\rho})(s^*y+sy^*)=|s|^2+\frac{\rho}{(1+\rho)^2}|y|^2-2\frac{\sqrt{\rho}}{1+\rho}\Re[s^*y]$$

Where the constants can be taken out of the expectancy and

- $\mathbb{E}[|s|^2]=1$
- $\mathbb{E}[|y|^2]=1+\rho$ *since y is sum of two gaussian*
- $\mathbb{E}[s^*y]=\sqrt{\rho}$ *since* $s^*y=s^*(\sqrt{\rho}s+z)=\sqrt{\rho}s^*s+s^*z$
  And now the expression simplifies to

$$MMSE(\rho)=1+\frac{\rho}{1+\rho}-2\frac{\rho}{1+\rho}=\frac{1}{1+\rho}$$

Example 1.28 (Verify Bias of 1.27).

To see the bias, select a specific value of the distribution:

$$\mathbb{E}[\hat{s}(y)|s=s] = \mathbb{E}\left[\frac{\sqrt{\rho}}{1+\rho}(\sqrt{\rho}s+z)|s=s\right]$$

So s becomes a constant and therefore

$$\mathbb{E}\left[\frac{\rho}{1+\rho}s + \frac{\sqrt{\rho}}{1+\rho}z\right] = \frac{\rho}{1+\rho}s = s - \frac{1}{1+\rho}s \neq s$$

the bias is therefore $-\frac{s}{1+\rho}$

This example shows the MMSe for a complex gaussian vector:

Example 1.31 (MMSE estimation of Complex Gaussian Vector TODO).

Let $s \sim \mathcal{N}_{\mathbb{C}}(0, R)$ *and* $z \sim \mathcal{N}_{\mathbb{C}}(0, I)$, *with* $Y = \sqrt{\rho}As + z$, *then*

$$\hat{s}(y) = \sqrt{\rho}RA^*(I + \rho ARA^*)^{-1}y$$

And it is possible to compute

$$\mathbf{E} = (R^{-1} + \rho A^* A)^{-1}$$

I-MMSE Relationship

Consider again $y = \sqrt{\rho}s + z$ with s independent from $z \sim \mathcal{N}_{\mathbb{C}}(0, 1)$.
Recall $I(\rho) = I(s; \sqrt{\rho}s + z)$, then it's derivative has significance as

$$\frac{1}{\log_2 e}\frac{d}{d\rho}I(\rho) = MMSE(\rho)$$

And also the integral form:

$$\frac{1}{\log_2 e}I(\rho) = \int_0^\rho MMSE(\xi)d\xi$$

This can be generalized to vectors
Let $\mathbf{y} = \sqrt{\rho}\mathbf{A}\mathbf{s} + \mathbf{z}$ with s independent from $\mathbf{z} \sim \mathcal{N}_{\mathbb{C}}(0, I)$

$$\frac{1}{\log_2 e}\nabla_A I(s; \sqrt{\rho}As + z) = \rho\mathbf{A}\mathbf{E}$$

Examples

In fact consider this example

Example 1.32 (I-MMSE for Complex Gaussian Scalar).

This is straight forward by recalling that for a complex gaussian scalar

$$I(\rho) = \log_2(1+\rho) = \frac{\ln(1+\rho)}{\ln 2} \overset{d/d\rho}{=} \frac{1}{\ln 2}\frac{1}{1+\rho}$$

By recalling that $\ln 2 \cdot \log_2 e = 1$ *the I-MMSE holds*.

And the following for vectors

Example 1.35 (I-MMSE for Complex Gaussian Vector)**.**

TODO

LMMSE Estimation

TODO